

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO4: Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards

(BL2, BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5

Total Marks:25

Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as

more devices are added due to increased data traffic on the shared medium.

2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.

3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.

4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.

5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than

star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRIC S FOR CALCU LATION Criteria	Weighta ge	Exempla ry (4)	Proficien t (3)	Developi ng (2)	Beginnin g (1)
Understa nding of Concepts	15%	Demonstr ates complete and in- depth understan ding of all concepts being compared	Good understan ding with minor gaps	Basic understan ding; some concepts unclear	Poor understan ding of concepts
Comparis on Depth	20%	Provides detailed, insightful , and meaningf ul comparis ons	Adequate comparis on with relevant points	Limited comparis on; lacks depth	Minimal or incorrect comparis on

		across multiple aspects			
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured

Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong justification	Clear conclusion	Basic conclusion	No or weak conclusion

Q1. Star Topology vs Bus Topology

Star topology connects every node individually to a central device such as a switch or hub, while bus topology connects all nodes to a single shared backbone cable. This structural difference drives most of the practical trade-offs between the two.

Cost and Installation: Star topology is generally more expensive to install because it needs a central device plus a dedicated cable run for every node, increasing both hardware and cabling cost. Bus topology, in contrast, uses one shared backbone, so it needs far less cable and no central device, making it cheap and quick to set up — which is why it was popular for small, low-budget networks.

Reliability and Fault Isolation: Star topology is more reliable at the node level: if one cable or device fails, only that node is affected and the rest of the network keeps working. Bus topology has the opposite problem — a break anywhere on the backbone, or a faulty terminator, can bring down the entire segment, since all traffic depends on that single cable.

Performance: Star topology performs better under load because the central switch manages traffic and can send data to only the intended recipient, minimizing collisions. Bus topology shares the same medium among all devices, so as more nodes are added, collisions and contention increase and performance degrades noticeably.

Scalability: Star topology scales more predictably — new nodes just plug into a free port on the switch. Bus topology has practical limits on cable length and the number of devices before signal degradation and collisions become unmanageable.

Conclusion: *Bus topology wins on raw cost and simplicity, but star topology is the stronger choice wherever reliability, performance, and future growth matter — which is why star (via Ethernet switches) dominates modern LANs, while bus is largely legacy.*

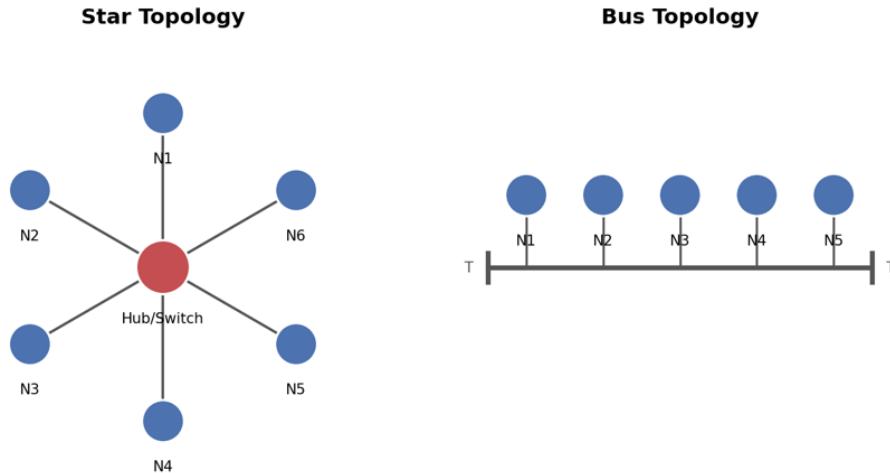


Fig. Q1 — Star Topology vs Bus Topology

Q2. Mesh Topology vs Star Topology

Mesh topology connects every node directly to every other node (or to many others in a partial mesh), while star topology routes all communication through one central device.

Fault Tolerance: This is the defining difference. Mesh topology offers very high fault tolerance because multiple paths exist between any two nodes — if one link fails, traffic simply reroutes through an alternate path. Star topology has only moderate fault tolerance: individual node failures are isolated, but the central hub/switch is a single point of failure, and its failure takes down the entire network.

Cost and Cabling: Mesh topology is far more expensive because the number of links grows rapidly with the number of nodes ($n(n-1)/2$ for a full mesh), requiring extensive cabling and many ports per device. Star topology needs only one cable per node plus the central device, making it much cheaper.

Complexity and Scalability: Star topology is simple to design, install, and expand — just add a node to a free switch port. Mesh topology becomes increasingly complex and costly to scale, since every new node must be wired to some or all existing nodes.

Use Case Suitability: Mesh topology's redundancy makes it ideal for critical infrastructure (e.g., core router backbones, WAN links, financial or military networks) where downtime is unacceptable. Star topology suits everyday LANs (offices, homes, classrooms) where cost and simplicity matter more than absolute fault tolerance.

Conclusion: *Mesh topology is the more robust and reliable design for mission-critical environments, but its cost and complexity make star topology the more practical everyday choice; the decision ultimately depends on how much redundancy the criticality of the network justifies.*

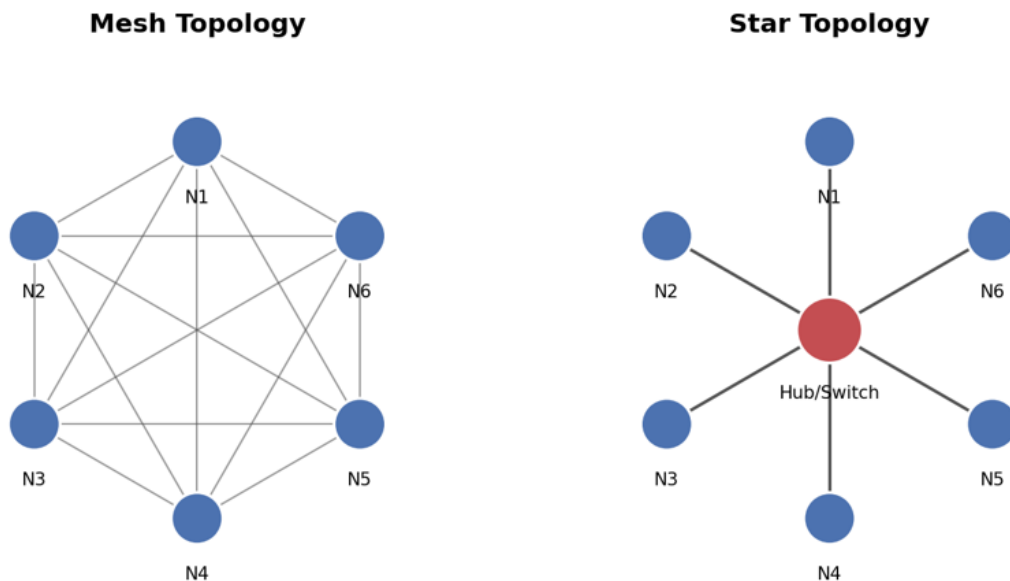


Fig. Q2 — Mesh Topology vs Star Topology

Q3. Bus Topology vs Mesh Topology

Bus topology relies on a single shared backbone cable to which all devices connect, whereas mesh topology requires a dedicated link between every pair (or many pairs) of connected nodes.

Installation and Simplicity: Bus topology is simple and fast to install — minimal cabling, minimal hardware, and straightforward for small networks. Mesh topology is highly complex to install because every node needs its own cable and port to reach other nodes, and this wiring effort grows sharply as more devices are added.

Cost: Bus topology is low-cost due to its single-cable design. Mesh topology is one of the most expensive topologies to implement because of the large number of cables, network interfaces, and ports required.

Reliability and Fault Tolerance: Here the comparison reverses — bus topology is fragile, since a single break in the backbone can disable the whole network, and locating that fault is difficult. Mesh topology is extremely fault-tolerant, since redundant paths mean a single link failure rarely disrupts communication.

Scalability: Bus topology has limited scalability due to signal degradation and cable-length restrictions as more devices are added. Mesh topology, while technically scalable, becomes prohibitively expensive and hard to manage at scale, so pure full-mesh designs are rarely used beyond small, critical cores.

Conclusion: *Bus topology trades reliability for simplicity and low cost, while mesh topology trades cost and installation complexity for maximum redundancy — they sit at opposite ends of the cost-versus-resilience spectrum, and the right choice depends on whether the network prioritizes affordability or fault tolerance.*

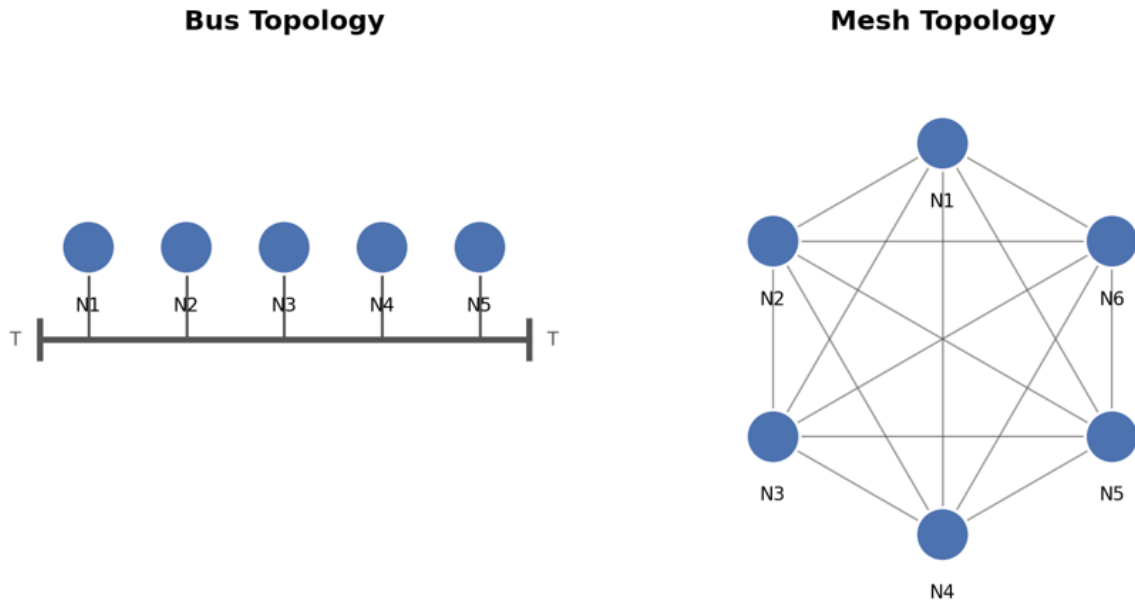


Fig. Q3 — Bus Topology vs Mesh Topology

Q4. Hybrid Topology vs Star Topology

Hybrid topology combines two or more basic topologies (e.g., star-bus, star-mesh) into a single network, while star topology uses only the single central-hub design.

Scalability: Hybrid topology is highly scalable — new segments or entire sub-topologies can be added without disrupting the existing structure, since different parts of the network can use whatever topology suits that segment's needs. Star topology is scalable only up to the port capacity of its central device; once that limit is reached, expansion requires additional switches or hubs, and eventually a hybrid design forms anyway (e.g., an extended star).

Flexibility and Design: Hybrid topology is flexible because it lets network designers pick the best topology for each part of the network —

for instance, star topology within departments and mesh topology for a resilient core/backbone. Star topology offers no such flexibility; it is uniform throughout.

Cost: Star topology is generally cheaper for small, single-segment networks, since it needs only one central device. Hybrid topology can cost more overall because it combines multiple devices and cabling schemes, though it can be cost-optimized by using expensive topologies (like mesh) only where resilience is truly needed.

Complexity: Star topology is simple to design and manage. Hybrid topology is inherently more complex, since it combines different topologies and their respective devices, requiring more careful planning and skilled administration.

Conclusion: *Star topology suits small, single-purpose networks where simplicity and lower cost matter, whereas hybrid topology is the better choice for larger organizations that need the network to grow flexibly and combine the strengths of multiple topologies — which is why most real-world enterprise networks are, in practice, hybrid.*

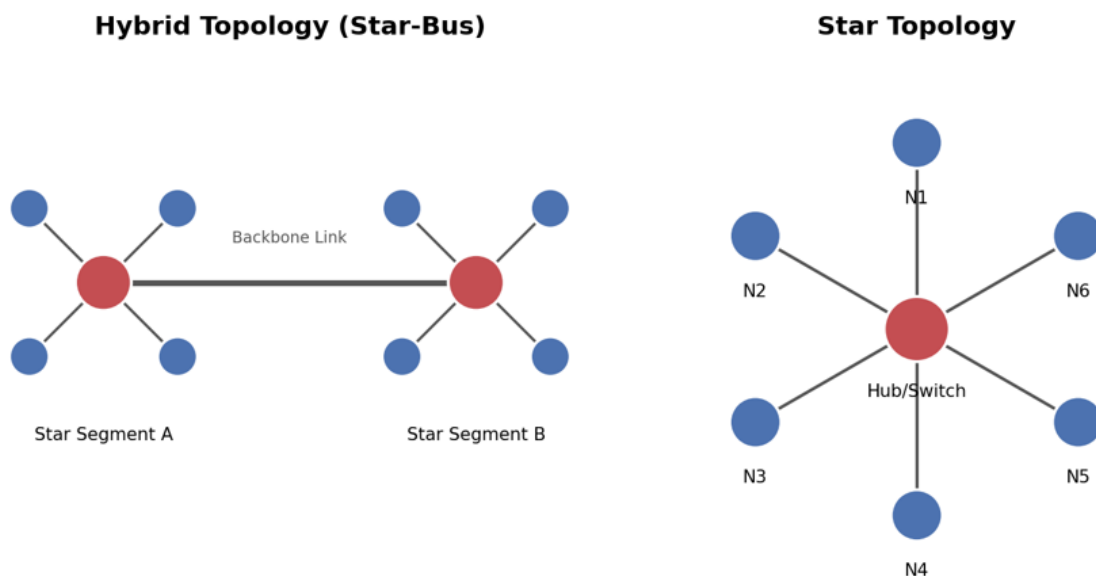


Fig. Q4 — Hybrid Topology vs Star Topology

Q5. Maintenance Comparison — Star, Mesh, Bus, and Hybrid

Star Topology: Easiest to maintain. Because each node has its own dedicated link to the central device, faults can be quickly isolated to a specific cable or node without affecting the rest of the network, and troubleshooting tools can test one segment at a time.

Bus Topology: Harder to maintain than star. Since all devices share one backbone cable, a fault (a break, a bad terminator, a loose connector) can disrupt the entire network, and locating the exact fault point along the shared cable is time-consuming, especially in longer runs.

Mesh Topology: Reliable in operation but difficult to maintain structurally. The large number of interconnections means there is a lot to monitor, and tracing, testing, or replacing any specific link is complex due to the sheer density of wiring — even though the network keeps functioning during a fault thanks to redundant paths.

Hybrid Topology: Moderate maintenance complexity. It benefits from the flexibility and partial fault isolation of its component topologies, but administrators must monitor and manage multiple combined structures simultaneously, requiring more skilled, coordinated administration than a single, uniform topology.

Ease-of-maintenance ranking (easiest → hardest): Star → Hybrid → Bus → Mesh

Conclusion: *Ease of maintenance is closely tied to how localized a fault's impact is and how quickly it can be traced. Star topology's centralized, node-isolated design makes it the easiest to manage, while mesh topology's dense interconnections — despite being the most fault-tolerant in operation — make it the hardest to maintain administratively. Hybrid topology occupies a middle ground, balancing flexibility against the overhead of managing multiple topology types at once.*

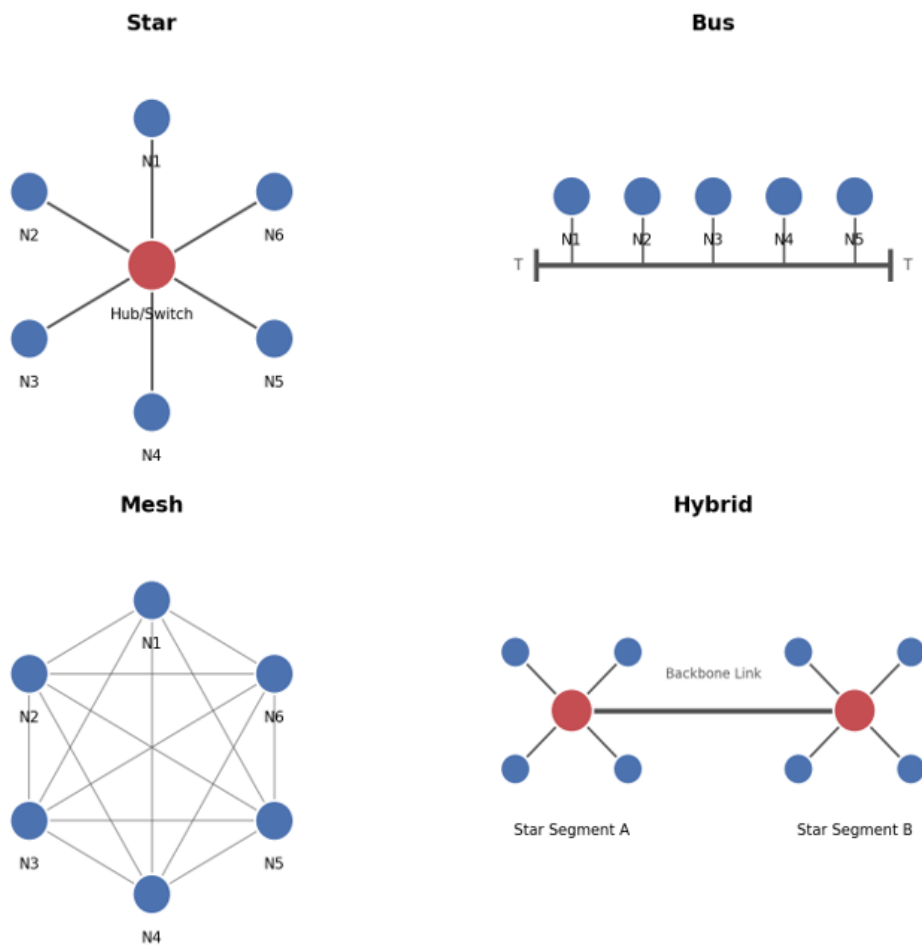


Fig. Q5a — Star, Bus, Mesh, and Hybrid Topologies

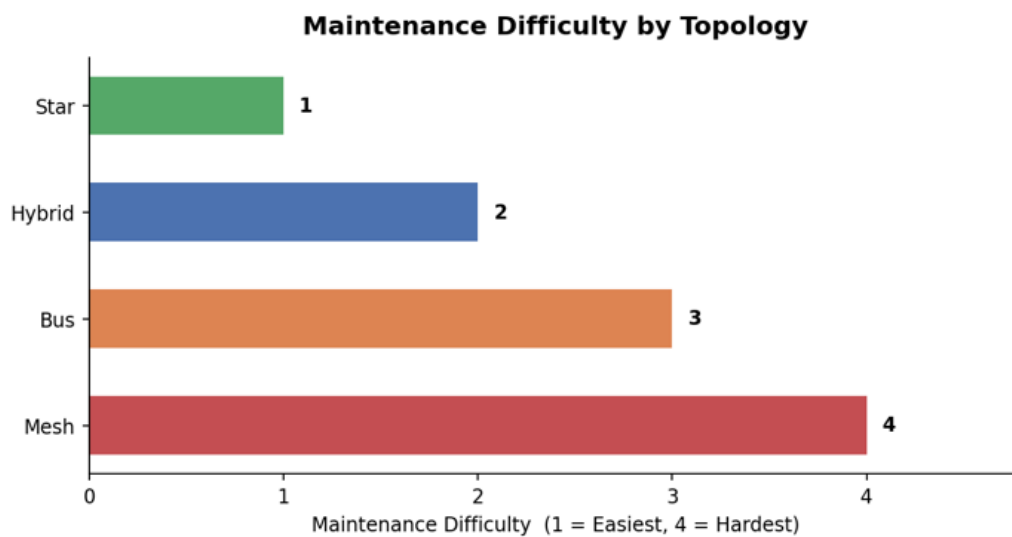


Fig. Q5b — Maintenance Difficulty Ranking (Star → Hybrid → Bus → Mesh)